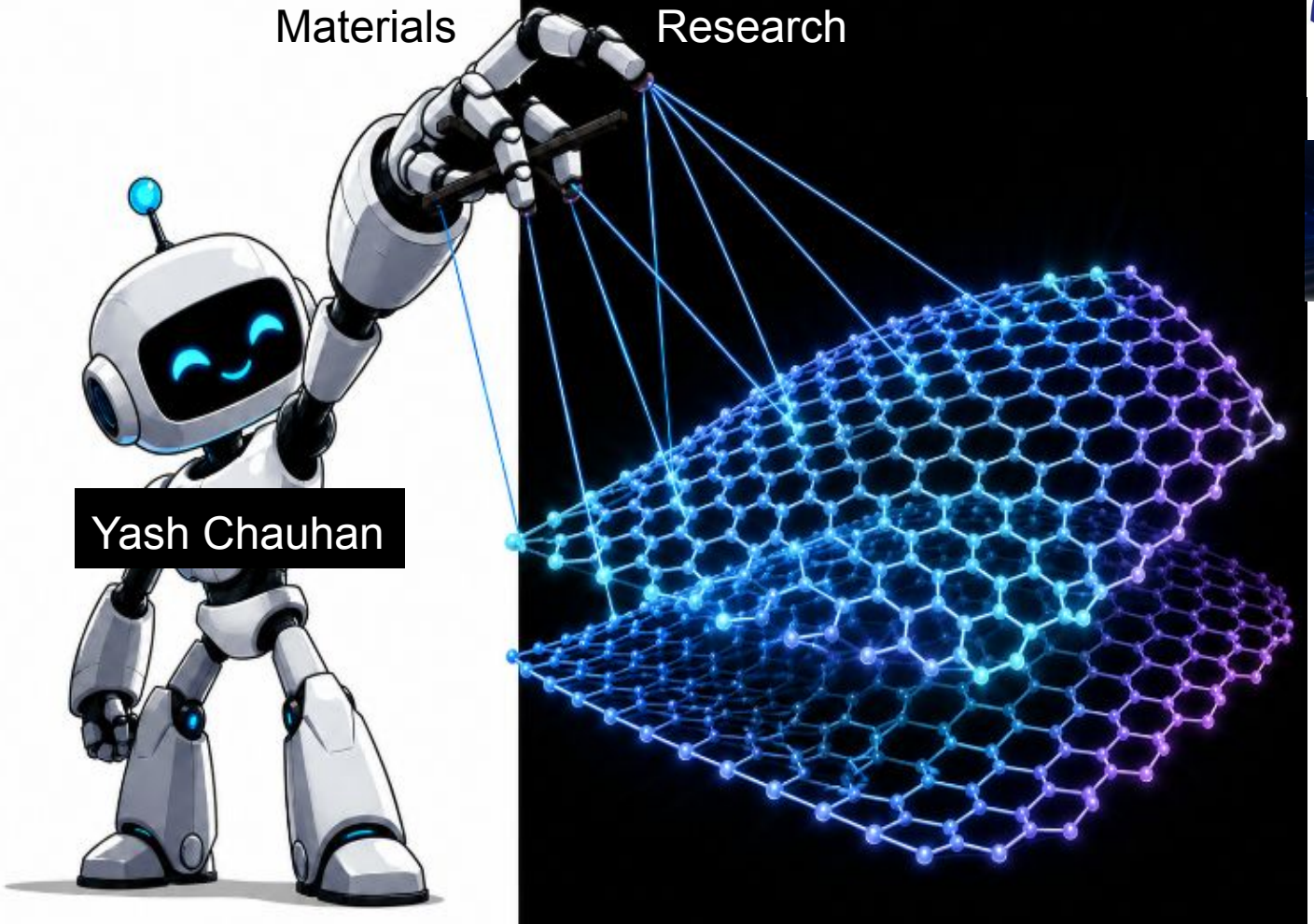
Fabrication, Devices, and Automated Workflows in 2D Materials Research



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LAM



राष्ट्रीय विज्ञान शिक्षा एवं अनुसंधान संस्थान
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Outline

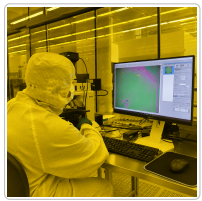
1. Overview
   - Timeline
   - Motivation
   - Why 2D materials?
2. Twisted MoS_2 Devices
   - Why twist 2D materials?
   - 2D device fabrication workflow
   - Target device
3. From Manual Fabrication to Automation
   - Automation for Twistronics
   - Newton Ring Problem
   - Pickup Detection
4. Conclusion

Section 1. Overview

Breakdown of the visit

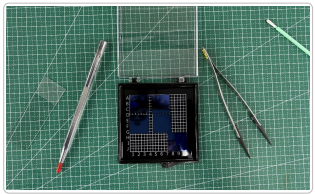
17 May – 27 May

- Si Cutting, Exfoliation
- Optical Microscope
- Basics of stacking



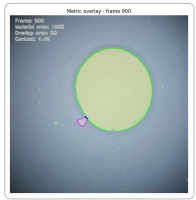
4 June – 11 June

- Trial Complete Stack
- Pickup detection

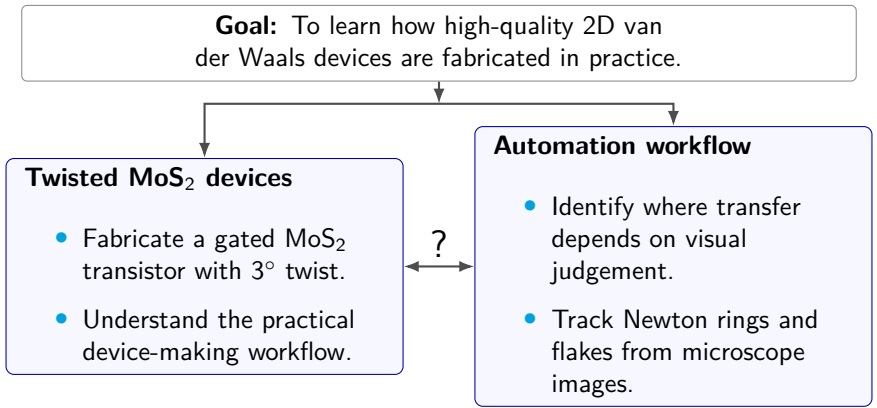


28 May – 3 June

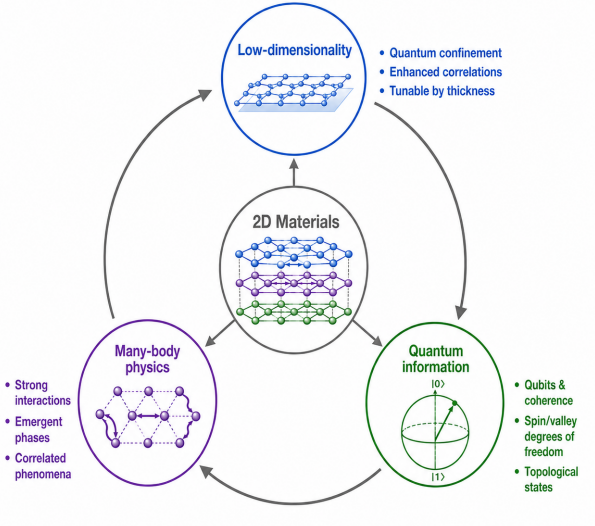
- Twisted MoS_2 device plan
- Cleanroom - EBL, EBE
- Newton-ring tracking



Motivation


Goal: To learn how high-quality 2D van der Waals devices are fabricated in practice.
Twisted MoS_2 devices
Fabricate a gated MoS_2 transistor with 3° twist.
Understand the practical device-making workflow.
?
Automation workflow
Identify where transfer depends on visual judgement.
Track Newton rings and flakes from microscope images.

Why 2D materials?

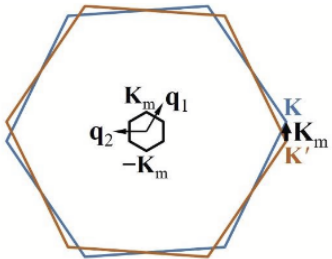


Why are they interesting?

- Reduced dimensionality enhances the role of interactions, confinement, and tunability.
- Van der Waals platforms allow clean stacking, twist engineering, and heterostructure design.

Section 2. Twisted MoS_2 Devices

Why twist 2D materials?



Relative rotation shifts the Brillouin-zone corners and defines a moiré wavevector.

Twist angle becomes a geometric knob for electronic structure.

A new length scale appears

A small relative angle θ produces a moiré period much larger than the atomic lattice constant:

$$L_m \simeq \frac{a}{2\sin(\theta/2)}$$

This long-period modulation can reshape low-energy bands by:

- folding bands into a mini-Brillouin zone,
- narrowing bandwidths,
- enhancing interaction effects,
- enabling gate-tuned moiré filling.

The Norwegian Academy of Science of Letters has decided to award the Kavli Prize in Nanoscience 2026 to

for foundational work that established the field of Twistronics



Eva Y. Andrei

Rutgers University, USA

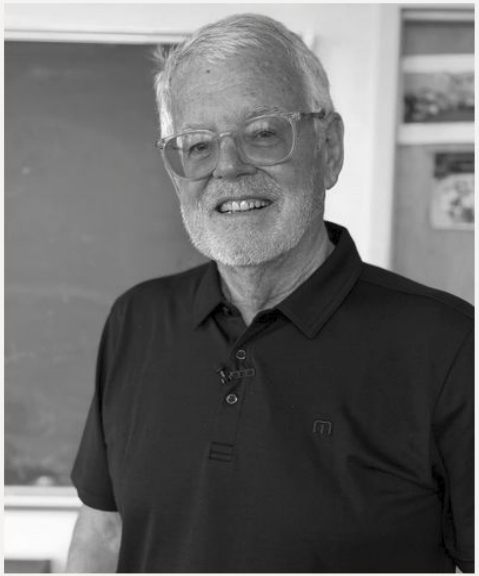
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Pablo Jarillo-Herrero

Massachusetts Institute of Technology, USA

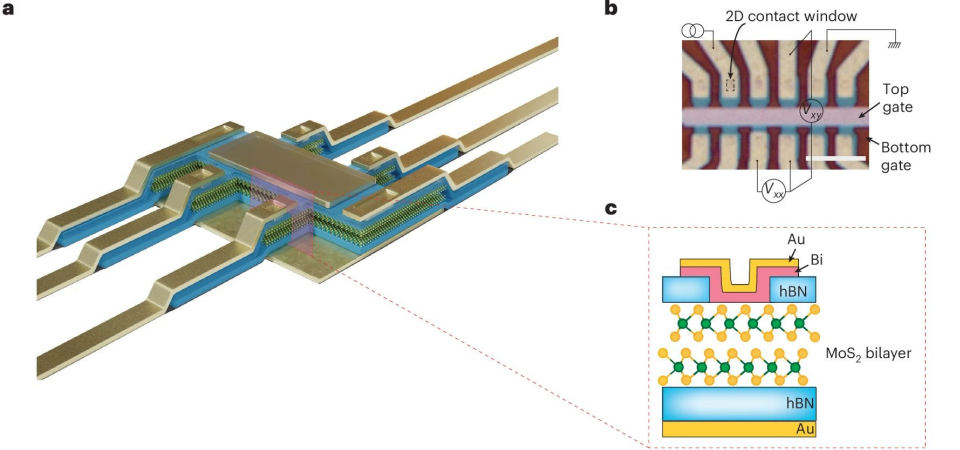
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Allan H. MacDonald

The University of Texas at Austin, USA

Read the full bio →

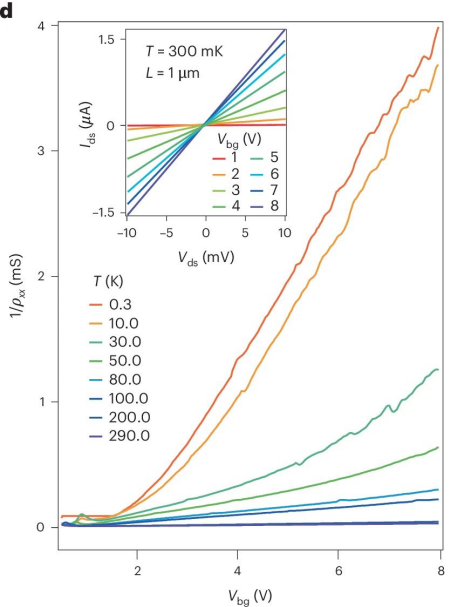


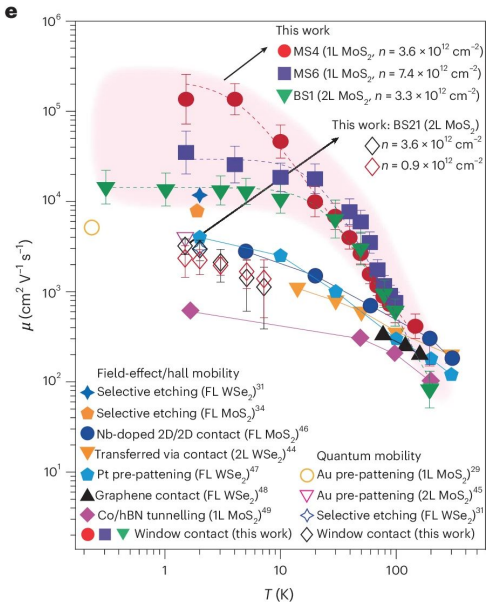
Contact problem in MoS_2

- Poor metal/MoS_2 contacts can mask intrinsic transport.
- Schottky barriers become severe at low temperature.

Window-contact solution

- hBN protects the MoS_2 channel.
- Etched windows expose only the contact regions.
- Metal contacts are formed without damaging the full channel.

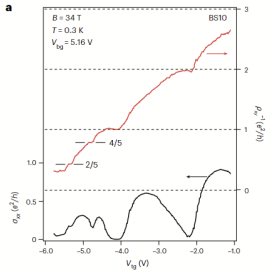




Ohmic low-T contacts + mobility > 10^5 cm^2 V^{-1} s^{-1} → MoS_2 becomes suitable for quantum transport experiments.

Zhao, S., et al. (2024).Fractional quantum hall phases in high-mobility n-type molybdenum disulfide transistors. Nature Electronics, 7(12), 1117–1125.

What high-quality MoS_2 devices can access

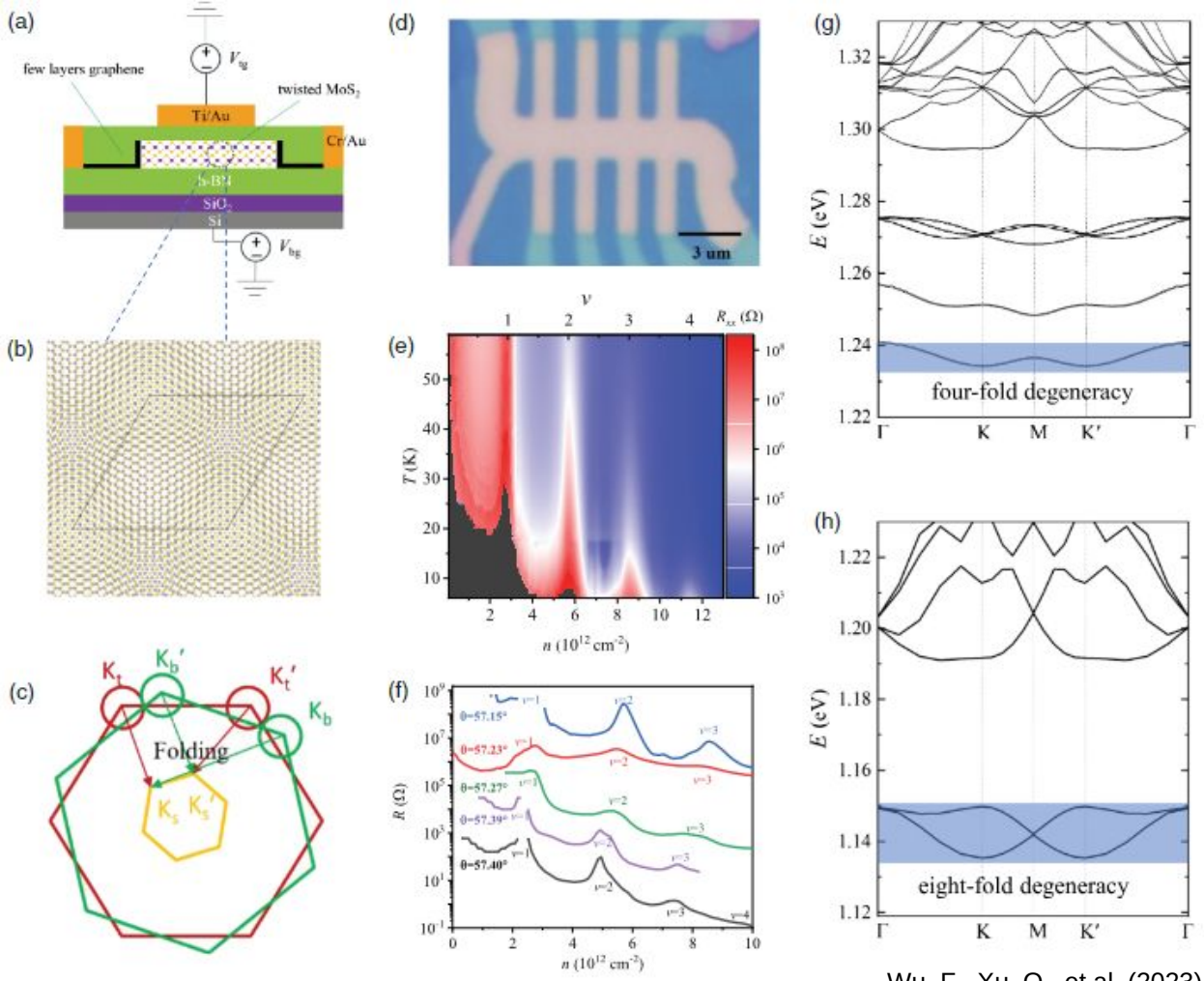


Fractional quantum Hall features in bilayer MoS_2.

Clean devices reveal many-body transport

$$\nu = \frac{nh}{eB} \qquad \nu_{\mathrm{int}} = 1, 2, 3, \ldots \qquad \nu_{\mathrm{frac}} = \frac{4}{5}, \frac{2}{5}$$

- Large B quantizes electron motion into Landau levels.
- Integer quantum Hall states mainly reflect single-particle quantization.
- Fractional states appear only when interactions dominate over disorder.
- Observing these features requires both a clean channel and low-resistance contacts.



1. hBN-encapsulated twisted bilayer MoS_2 is used as a clean dual-gated moiré device.
2. Moiré lattice: Near 57.5° twist, the two MoS_2 layers form an AB-stacked moiré superlattice.
3. Resistance peaks appear when the moiré band is filled by an integer number of electrons.
4. These peaks indicate insulating states caused by electron–electron interactions.
5. The calculated moiré minibands are narrow, so the interaction scale becomes large compared with the bandwidth: $U / W \gg 1$

Wu, F., Xu, Q., et al. (2023).Giant correlated gap and possible room-temperature correlated states in twisted bilayer MoS2. Phys. Rev. Lett., 131, 256201.

Why study twisted MoS_2 near 3°?

Twisted MoS_2 is already known to host correlated moiré states.
The next question is how these states evolve in a different twist-angle regime.

$$L_m \simeq \frac{a}{2\sin(\theta/2)}, \qquad a_{\mathrm{MoS_2}} \simeq 0.315\ \mathrm{nm}, \qquad \theta = 3^\circ \quad \Rightarrow \quad \boxed{L_m \approx 6\ \mathrm{nm}}$$

Starting point: coupled twisted bilayer

$$H = \begin{pmatrix} H_1 & \mathcal{T}(\theta, \mathbf{r}) \\ \mathcal{T}^\dagger(\theta, \mathbf{r}) & H_2 \end{pmatrix}$$

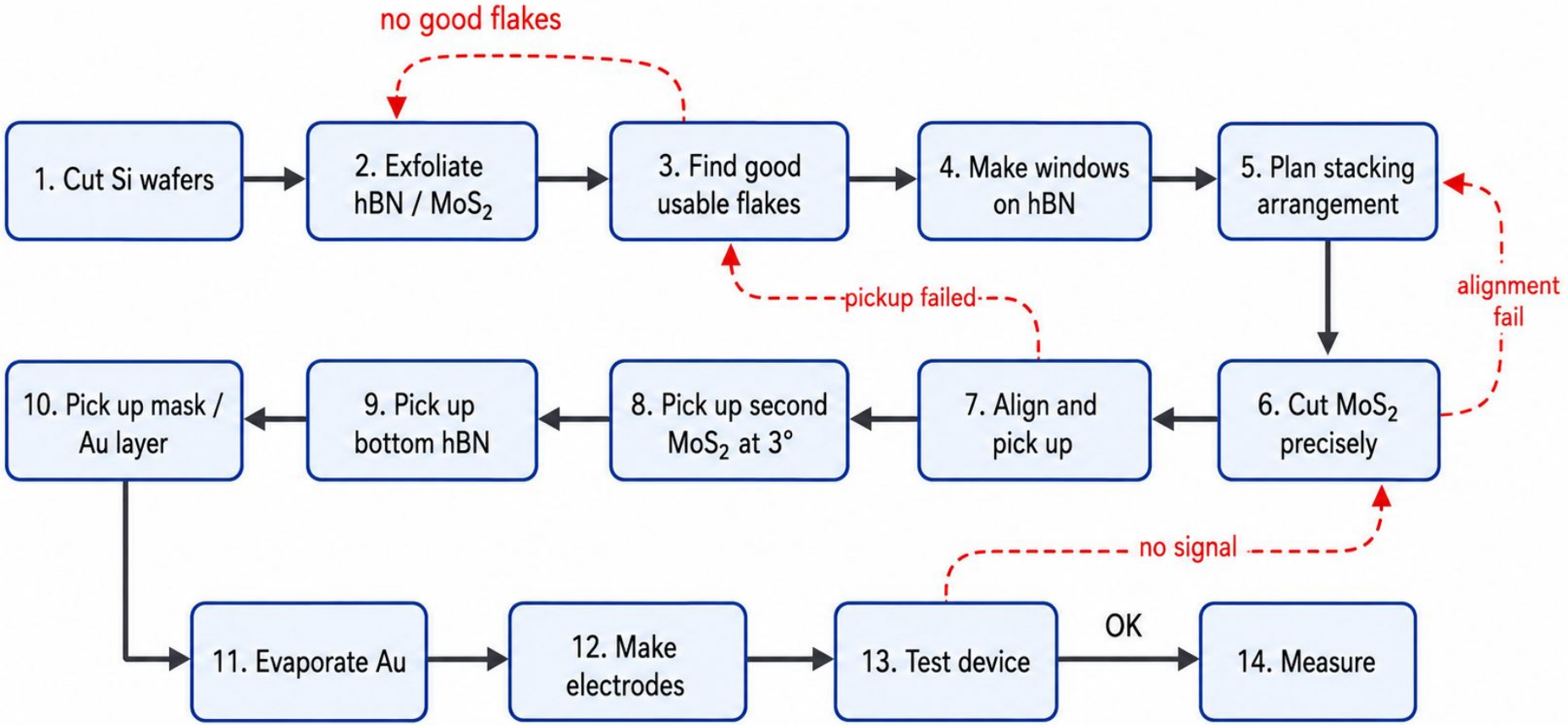
$\mathcal{T}(\theta, \mathbf{r}) \Rightarrow$ moiré potential $\Rightarrow$ minibands

Here H_1, H_2 describe the two MoS_2 layers, while $\mathcal{T}$ encodes twist-dependent interlayer coupling.

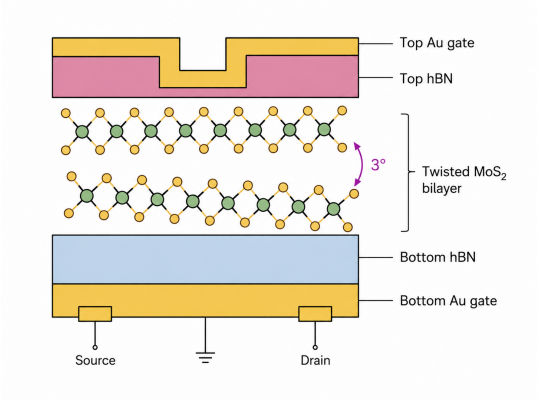
What changes at 3°?

$$\theta \to L_m \to W, \qquad \frac{U}{W}$$

- shorter moiré period than smaller-angle devices
- modified miniband bandwidth W
- different charge localization pattern
- possible change in correlated phases


no good flakes
1. Cut Si wafers
2. Exfoliate hBN / MoS_2
3. Find good usable flakes
4. Make windows on hBN
5. Plan stacking arrangement
pickup failed
alignment fail
10. Pick up mask / Au layer
9. Pick up bottom hBN
8. Pick up second MoS_2 at 3°
7. Align and pick up
6. Cut MoS_2 precisely
no signal
11. Evaporate Au
12. Make electrodes
13. Test device
OK
14. Measure

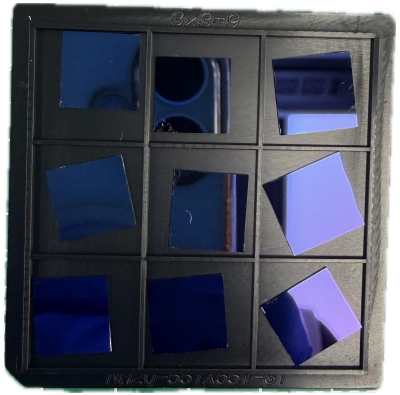
Target device: dual-gated twisted MoS_2 heterostructure



$\mathrm{Au/hBN/tMoS_2/hBN/Au}$

Design goals

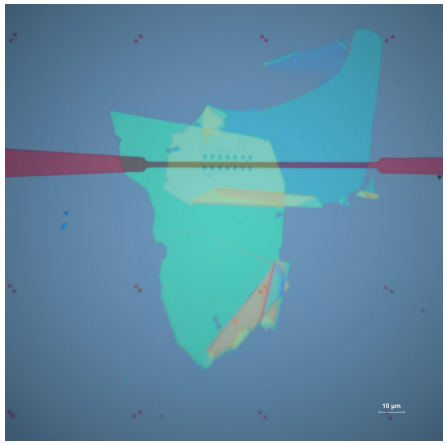
- 3° twisted MoS_2 active channel
- hBN-encapsulated clean interfaces
- dual-gate electrostatic control
- source–drain transport readout

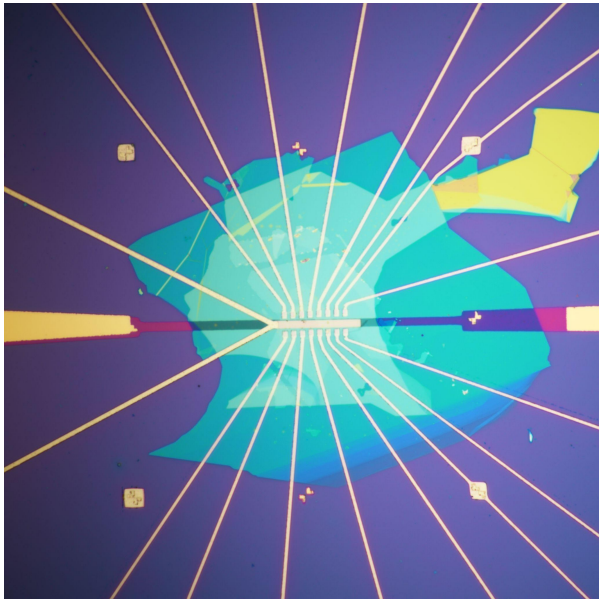

3x3=9


10 µm




10 µm





Challenges encountered

Flake selection and preparation

- Large, clean hBN and MoS_2 flakes were limited.
- Suitable flakes had to satisfy both size and cleanliness requirements.
- Window-patterned hBN added extra alignment constraints.

Twisted-stack assembly

- Preparing a controlled $\sim 3^\circ$ twist was more delicate than a standard stack.
- Flakes could shift during pickup or transfer.

Pickup and release

- PC stamp behaviour was difficult to control reproducibly.
- Some steps suffered from incomplete pickup or release.
- Small changes in contact, temperature, or timing affected success.

Main bottleneck

- Many decisions depended on real-time visual judgement.
- A complete device was therefore not finished within the available time.

In 2D device fabrication, small human judgement errors can cause large delays.

manual visual feedback

$\Longrightarrow$

missed pickup, misalignment, incomplete release

$\Longrightarrow$

repeating the workflow

Section 3. From Manual Fabrication to Automation

AI-driven robotic stacking for twistronics

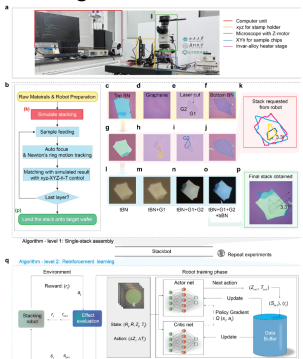
Why automate stacking?

- Twistronic devices require precise layer pickup, alignment, and release.
- Manual stacking is slow, variable, and difficult to scale.
- The robot converts dry transfer into an image-guided workflow.

Core idea

Microscope images are used not only for observation, but also for controlling the stacking process.

AI-assisted workflow for flake recognition, planning, pickup, and stacking.



Computer vision + autofocus + Newton-ring tracking + coordinated hardware control.

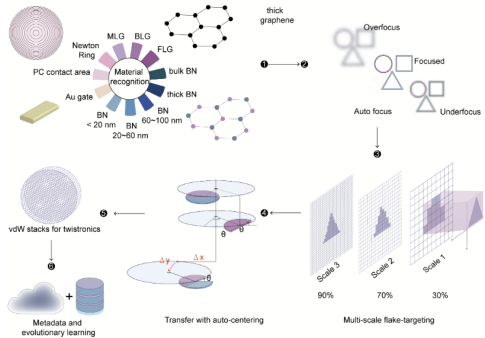
Newton rings as real-time contact feedback

What is being tracked?

- During pickup, the PC/PDMS stamp forms a contact region.
- This appears optically as a Newton-ring / wetting-front pattern.
- Its motion tells us how the stamp is approaching, wetting, and retracting.

$$R(t), \quad A(t), \quad \dot{R}(t)$$

radius, area, and growth rate become quantitative feedback signals.



Newton-ring tracing during the robotic transfer loop.

Li, X., He, J., Liu, H., et al. (2026). An ai-driven robotic system for two-dimensional hetero-assemblies. https://arxiv.org/abs/2605.20420

From manual prompting to automatic contact detection

Goal: remove the human from the first detection step

Current bottleneck

- The contact/ring region is often initialized manually.
- Tracking quality depends on the first prompt.
- This limits direct use in live robotic feedback.

Desired pipeline

- Detect contact directly from microscope frames.
- Segment the ring/contact contour automatically.
- Extract radius, area, contour, and growth rate.

$$I_t(x, y) \longrightarrow M_t^{\text{contact}}(x, y) \longrightarrow \{A_t,\ R_t,\ \dot{R}_t,\ C_t\}$$

frame $\rightarrow$ contact mask $\rightarrow$ quantitative signal for feedback control

Contact-front detection as image segmentation

The Newton-ring/contact region is treated as a color-based segmentation problem.

$$I_t(x, y) = (R_t, G_t, B_t) \quad \longrightarrow \quad (H_t, S_t, V_t)$$

Candidate contact pixels are selected using HSV thresholds:

$$M_t(x, y) = \mathbf{1}(H_{\min} < H_t < H_{\max})\mathbf{1}(S_t > S_{\min})\mathbf{1}(V_t > V_{\min})$$

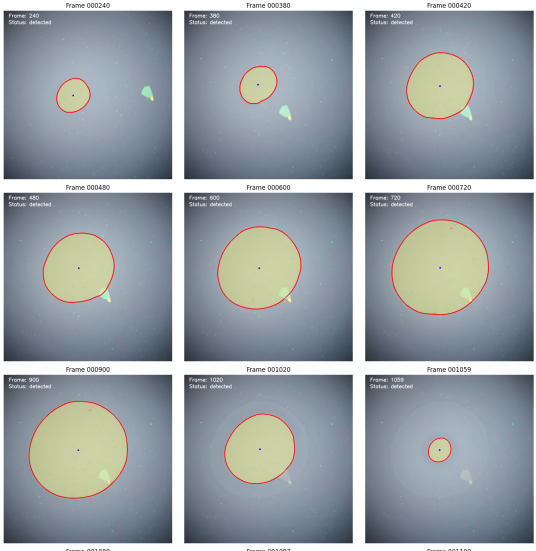
Mask cleanup

- remove isolated noisy pixels
- fill small holes inside the detected region
- keep only connected regions with reasonable size

Output

A cleaned candidate mask for the Newton-ring/contact region.

Tracking the physical contact front



Contour selection

The mask may contain several regions:

$$M_t = \bigcup_i \Omega_i$$

Choose the region using:

$$A_i, \qquad \chi_i = \frac{4\pi A_i}{P_i^2}, \qquad c_i$$

area, circularity, and center position

Tracked outputs

$$R_t \sim \sqrt{\frac{A_t}{\pi}}, \qquad v_t = \frac{dR_t}{dt}$$

Current status: promising but not yet reliable

The video demonstrates feasibility, not robustness.

What works

- contact regions can be detected automatically
- contours can be tracked frame-by-frame
- radius and growth-rate signals can be extracted

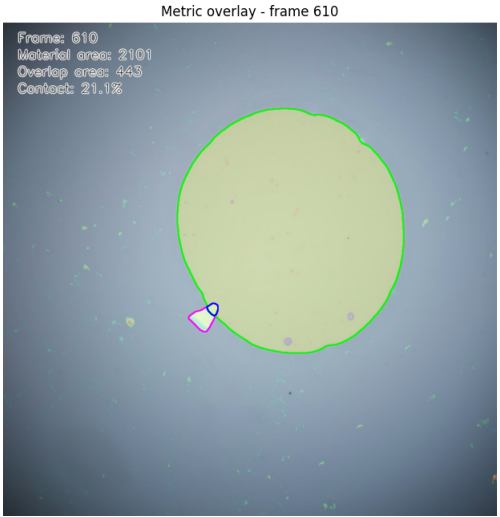
What remains unclear

- sensitivity to illumination and focus
- different Newton-ring color patterns
- bubbles, wrinkles, and flake edges
- transferability across many stacking videos

Takeaway

Color and contrast can be deceptive. The method must be validated on a much larger set of transfer experiments before being used as a reliable automation signal.

Contact tracking is not the same as pickup detection



Contact tracking

Is the PC touching the flake?

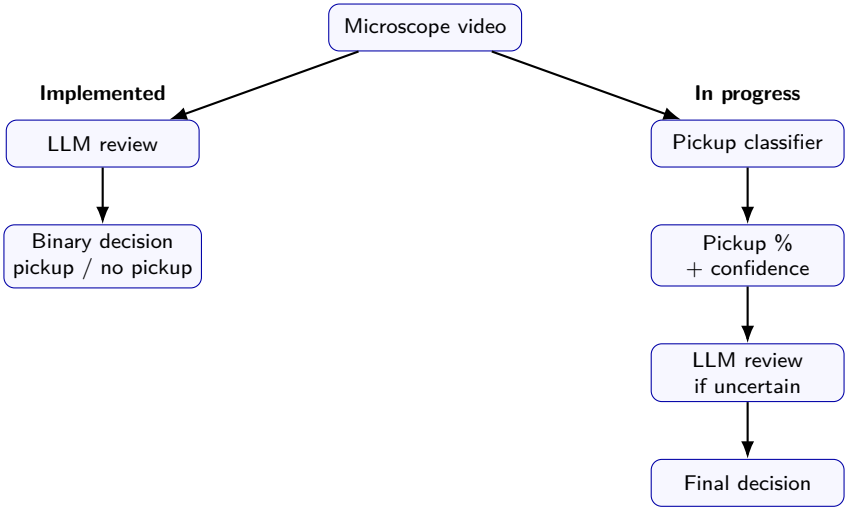
Pickup detection

Did the flake leave the substrate?

A simple pickup estimate is

$$P = 100\left(1 - \frac{\text{remaining area}}{\text{initial area}}\right)$$

Two routes toward pickup detection



Prototype status and real-time challenges

Current capabilities

- Newton-ring/contact tracking
- Contact-area estimation
- Pickup detection from post-retraction frames
- LLM-based review of transfer outcomes
- Video overlays and metric extraction

$$\text{Video} \rightarrow \text{Metrics} \rightarrow \text{Decision}$$

Challenges for real-time deployment

- changing illumination and microscope settings
- different flake colors and contrasts
- bubbles, wrinkles, and contamination
- partial pickup events
- robustness across different devices and operators

Section 4. Conclusion

Conclusion

Twisted MoS_2 devices

- High-quality MoS_2 devices now access correlated and moiré physics.
- A dual-gated 3° twisted MoS_2 stack was selected as the target device architecture.
- Fabrication challenges highlighted the importance of reliable transfer feedback.

Automation and computer vision

- Contact-front tracking was explored using image segmentation.
- Pickup detection was formulated as a separate post-retraction decision problem.
- Initial LLM-based and classifier-assisted approaches were developed.

Future direction

Integrate contact tracking and pickup detection into a real-time feedback layer for automated dry-transfer and robotic stacking.

Acknowledgements

I would like to sincerely thank Prof. Zheng Vitto Han for giving me this opportunity and for hosting me in his lab, even for a short visit.

I am especially grateful to Hui for taking the time to teach me the fabrication process and for being patient throughout the learning process.

I would also like to thank Jinkun for helping me get started with the automation project.

Finally, I am thankful to the entire group for making me feel welcome during my stay in China.

Section 5. References

- Li, X., He, J., Liu, H., et al. (2026). An ai-driven robotic system for two-dimensional hetero-assemblies. https://arxiv.org/abs/2605.20420
- Wu, F., Xu, Q., et al. (2023).Giant correlated gap and possible room-temperature correlated states in twisted bilayer $\mathrm{MoS2}$. **Phys. Rev. Lett.**, **131**, 256201. https://doi.org/10.1103/PhysRevLett.131.256201
- Zhao, S., et al. (2024).Fractional quantum hall phases in high-mobility n-type molybdenum disulfide transistors. **Nature Electronics**, **7**(12), 1117–1125.